



Footage Sourcing Guide — Vision Pipeline Inputs



Footage is the one artifact that cannot be authored as text. This guide gives the two zero-cost acquisition paths, the license rules, and the processing manifest. Budget: ~half a day of sourcing + clipping.

Path A (default per Ch. 19 §19.4): synthetic frames

Render frames **from the engine's own world state** — top-down map tiles with unit markers composited at true positions, plus noise. Zero licensing risk, perfectly labeled ground truth (you know every object's position), and the CV demo stays honest: the pipeline detects markers it was trained/configured for.

- Generator: small Python script (Pillow + staticmaps) — `tools/synth_frames.py`, Phase 4 sprint 1 task.
- Output: 720p PNG sequences, 2–5 fps, named `synb_<scenario>_<seq>.png`.
- This is the default demo path; real footage is the *stretch* path below.

Path B: free real aerial footage (for realism demos)

Source	License	Use for	Rules
Pexels (aerial/drone category)	Pexels license — free use, no attribution required	terrain flyovers, convoy-like road traffic, flood water	avoid clips with identifiable people/plates
Pixabay (videos)	Pixabay Content License — free use	same	same
Wikimedia Commons	CC-BY / CC-BY-SA per file	military exercise footage (public affairs releases)	attribution required — record in NOTICE
VisDrone dataset	academic/research only	fine-tuning experiments only	never redistribute; never in the public

Source	License	Use for	Rules
			demo (Ch. 19 §19.4)

Selection rules: rural/terrain-dominant clips; no identifiable faces or license plates; 10–60 s per clip is enough — the pipeline demos detection, not cinematography.

Processing (ffmpeg — free)

```
# normalize every sourced clip: 720p, 10 fps, strip audio,
h264
ffmpeg -i raw/clip.mp4 -vf "scale=1280:720,fps=10" -an -c:v
libx264 -crf 28 footage/recon-04.mp4
```

footage/manifest.yaml

```
# every file must have an entry – CI checks this against NO
TICE
- file: recon-01.mp4
  scenario: SC-2          # border ridge flyover
  source: pexels
  source_url: <fill on download>
  license: pexels
  attribution_required: false
- file: recon-02.mp4
  scenario: SC-3          # low-level UAV approach
  source: pixabay
  source_url: <fill>
  license: pixabay-content
  attribution_required: false
- file: recon-03.mp4
  scenario: SC-4          # mountain road / convoy corridor
  source: wikimedia
  source_url: <fill>
  license: CC-BY-4.0
  attribution_required: true # add to NOTICE
- file: recon-04.mp4
  scenario: SC-8          # flood-water aerial (HADR)
```

```
source: pexels
source_url: <fill>
license: pexels
attribution_required: false
- file: synb_sc2_*.png
  scenario: SC-2
  source: synthetic (tools/synth_frames.py)
  license: project (Apache-2.0)
  attribution_required: false
```

Acceptance checklist

- ☐ Every footage file has a manifest entry with a real `source_url`
- ☐ CC-BY items listed in `NOTICE`
- ☐ No VisDrone material in the repo or public demo
- ☐ Vision Review screen (S9) demo runs end-to-end on Path A alone — real footage is optional garnish, never a dependency